Evaluating moving target defence against advanced persistent threat attackers modelled on documented intrusion campaigns

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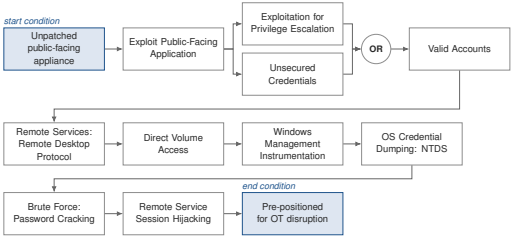
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MTD is proposed against APT attackers, but rarely tested against one

- **Advanced persistent threat (APT)**: one objective, one target, months or years [Alshamrani 2019]
- Volt Typhoon: at least five years inside US critical infrastructure; its reconnaissance stayed valid [CISA 2024]
- **Moving target defence (MTD)**: keep changing the network, so what the attacker learned goes out of date [Cho 2020]
- Yet MTD is tested against attackers far simpler than an APT [Jalowski 2026]

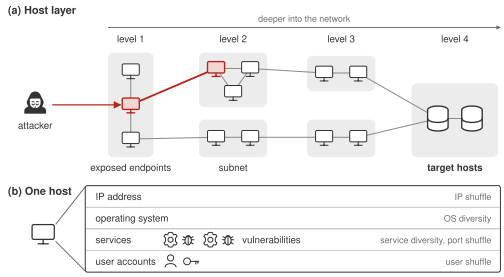


Volt Typhoon's intrusion as an **attack flow**: the steps an analyst recorded from the threat report, in order

Research question: How does MTD perform against APT attackers?

Previous work: MTDSim and its scripted baseline attacker

- **MTDSim**: UWA's MTD simulator [Brown 2023; Zhang 2023; Tay 2024]
- 50 hosts in four levels, from exposed endpoints to target hosts
- Seven **MTD mechanisms**, one deployed every **deployment interval** (200 s by default):
  - **host layer**: IP shuffle, two topology shuffles
  - **service layer**: port shuffle, OS and service diversity
  - **credentials**: user shuffle
- **Baseline attacker**: six **attack actions** in a fixed order



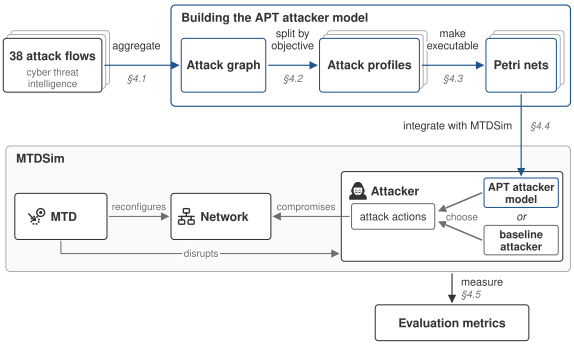
The gap: recent MTD attacker models have at most one APT property in full

	Property	Baseline attacker 2023	Masud et al. 2025	Kim et al. 2026
1	Campaign persistence	✓	part	part
2	Objective-specific behaviour	✗	✗	✗
3	Choice of attack strategy	part	✗	✗
4	Adaptivity to MTD disruption	part	✗	✗
5	Detection avoidance	✗	✗	part
6	Cost–benefit reasoning	✗	✗	✗
7	Learning from past deployments	✗	✗	✗
8	Deployment-strategy awareness	✗	✗	✗

1, 2, 4, 5: the APT definition [Alshamrani 2019; NIST 2011]. 3, 6, 7, 8: two MTD surveys [Cho 2020; Jalowski 2026].

Gap: how MTD performs against APT attackers is unmeasured.

Approach: build an APT attacker model from threat reports, run it in MTDSim

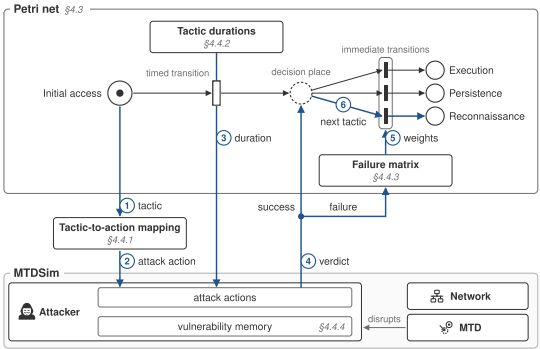


- **SQ1**: 38 attack flows [MITRE CTID] → an **attack graph** over the 15 ATT&CK **tactics** → four **attack profiles** by objective:

c1	stole data	19 flows
c2	impeded systems	7
c3	both	7
c4	neither	5

- **SQ2**: each attack profile runs in MTDSim as a Petri net
- **SQ3**: both attackers, same networks, same MTD

A Petri net picks the next tactic; MTDSim's attack actions carry it out



- Next tactic drawn by how often the attack flows took that step
- **Tactic-to-action mapping**: 8 of 15 tactics run an attack action; 7 are **unmapped** (e.g. Exfiltration) and only take time
- **Tactic duration**: up to 45 s on average
- **Failure matrix**: after a failure, reweights the next tactic (failed Initial Access → Reconnaissance: 0.06 becomes 0.75)
- Threat reports record no timing or failures: declared, then stress-tested

Experiment: both attackers, the same 1 000 networks, every MTD setting

Attacker	baseline attacker; APT attacker model on each of c1 to c4
MTD	none; each of the seven mechanisms alone; random and alternative rotation; MTDShield as released
Deployment interval	50, 100, 200, 500, 1 000 and 2 000 s
Run	targeted scenario: ends at the first target host compromised, or at 15 000 s
Seeds	1 000 per configuration; same seed, same network, for both attackers

- **NCR**, network compromise ratio: share of hosts compromised [Zhang 2023]
- **NCR reduction**
  $$= 1 - \frac{\text{NCR under MTD}}{\text{NCR with no MTD}}$$

- Each attacker against its own runs with no MTD
- MTD ranked by Scott–Knott effect-size test; rankings compared by Spearman's ρ

No MTD: the APT attacker model acts more like an APT, but compromises less

Property	Metric	APT attacker model (c1 to c4)	Baseline attacker
Choice of attack strategy	distinct openings (first 8 steps)	179 to 947	3
Detection avoidance	attack rate (attack actions per minute)	0.68 to 1.23	1.71
	attack confidentiality (share not flagged)	89 to 98 %	82 %
Objective-specific behaviour	steps in Exfiltration / Impact	c1, c3 / mostly c2, c3	neither
Attack outcome	ASP (runs reaching a target host)	0.04 to 0.14	0.66
	NCR	0.15 to 0.21	0.50

Its tactic durations and the order of its tactics cause both halves: ASP and NCR alone rate it the weaker attacker for what makes it APT-like.

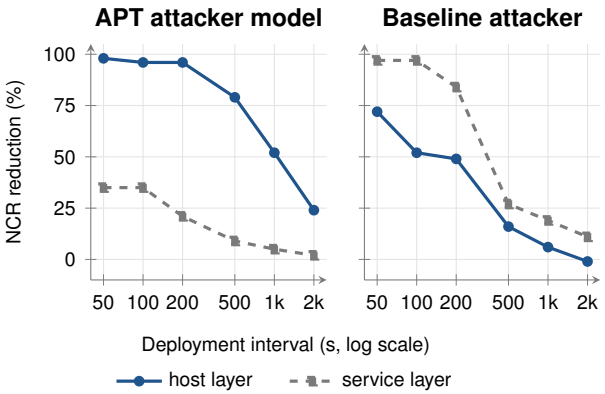
Headline: the most effective MTD mechanism differs between the two attackers

NCR reduction at 200 s	APT attacker model	Baseline attacker
host layer		
IP shuffle	**96 % (1)**	68 % (2)
complete topology	**95 % (1)**	42 % (3)
host topology	**95 % (1)**	38 % (3)
service layer		
service diversity	36 % (4)	**92 % (1)**
port shuffle	20 % (6)	**91 % (1)**
OS diversity	7 % (7)	68 % (2)
credentials		
user shuffle	−10 % (8)	−5 % (5)

(rank) among all ten MTD settings; APT attacker model averaged over c1 to c4.

- Rank 1 against the APT attacker model: **host layer** only
- Rank 1 against the baseline attacker: **service layer** only
- Service diversity: 92 % against 36 %, on the same networks
- Rankings uncorrelated: Spearman's $\rho = 0.01$

The difference holds at every deployment interval, and has a mechanism



Each line: the mean over the layer's MTD mechanisms

- The service layer disrupts only scan port, exploit and brute force
- The baseline attacker does these back to back; the APT attacker model sits mostly in unmapped tactics
- Service-layer deployments that block an attack action: 95 % against 15 %
- Each host-layer deployment costs the APT attacker model 242 to 324 s; the baseline attacker none

So what: the attacker model can decide which MTD looks best

MTDShield learned against the baseline attacker; it deploys service diversity in 70 % or more of its decisions.

NCR reduction, 200 s	Baseline	APT model
MTDShield	**68 %**	28 %
random deployment	33 %	**75 %**

- **Learned deployment strategies**: test against an attacker model they were not trained on, beside random deployment
- **Rankings**: report each with its attacker model; compare across studies only on a shared one
- **Cost**: to halve NCR, service diversity deploys every 1 000 s against the baseline attacker, every 100 s against the APT attacker model

An evaluation against one simplified attacker cannot tell you how MTD performs against an APT.

Limitations: what the result does not show

- **One simulator**: MTDSim's default 50-host network; deployments cost nothing
- **Six attack actions cap the model**: 7 of 15 tactics unmapped; no data theft, no impact, no IDS to evade
- **Declared values**: tactic durations and failure matrix by judgement; only the 45 s duration moves NCR much, and the ranking holds from a quarter to four times it
- **Small corpus**: 38 attack flows, 18 attributed to a known group; three attack profiles rest on 5 to 7
- **Cause not isolated**: tactic durations or the order of tactics; MTDSim's disruption rules were written for the baseline attacker

Conclusion: in MTDSim, the most effective MTD depends on the attacker model

- **Answer**: the host layer disrupts the APT attacker model most, the service layer the baseline attacker, at every deployment interval
- **Contribution 1**: an APT attacker model from 38 attack flows that any MTDSim study can run; 2 of 8 properties in full and 3 in part, against at most 1 in full for three recent models
- **Contribution 2**: evidence for evaluating MTD against attacker models grounded in documented behaviour
- **Future work**: attack actions for actions on objectives, and an IDS; separate tactic durations from the order of tactics; deployment cost; other networks; MTDShield retrained

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Questions? Code and experiments: github.com/marc-la/MTDSim

Backup: why the APT attacker model compromises less with no MTD

- Unmapped tactics take 34 to 57 % of an attack profile's run, and 15 to 30 % of its attack actions fail because their preconditions are missing (e.g. exploit before scan port)
- Rerun with the baseline attacker's times in place of its tactic durations, same order of tactics: ASP rises from 0.09 to 0.45, against the baseline attacker's 0.66, closing 63 % of the gap
- The order of its tactics most likely accounts for the rest
- The rankings compare each attacker with its own NCR with no MTD, so the lower starting point is unlikely to decide which mechanism ranks first

Dissertation Section 6.1, Appendix G.3, threat T2. Appendix G analyses were chosen after the results were seen (T9).

Backup: footholds under the host layer

- Under the three host-layer mechanisms at 50 to 200 s, 70 to 89 % of the APT attacker model's runs compromise no host, against 3 to 20 % of the baseline attacker's
- With no MTD: 1 % and none
- So the host layer disrupts the APT attacker model mostly *before* its first foothold: after a host-layer deployment it has no host until its Petri net next reaches Lateral Movement
- From 500 s the share falls for both attackers, and the host layer still ranks first against the APT attacker model

Dissertation Appendix G.4 (post hoc, T9); the claim stands on Chapter 5 without it.

Backup: are the declared tactic durations driving the result?

Mean duration (band)	low	declared	high
4.5 s (2.25 to 9)	0.165	0.163	0.159
35 s (17.5 to 70)	0.172	0.163	0.145
36 s (18 to 72)	0.170	0.163	0.148
45 s (11.25 to 180)	**0.267**	0.163	**0.059**

NCR with no MTD, each value moved alone, 400 runs per cell.

- Bands fixed before the run; one value moved at a time
- Only the 45 s (Persistence, Stealth, Command and Control) moves NCR much: Command and Control is the only tactic that finds new hosts
- At a quarter and four times the 45 s, rank 1 still shares no mechanism with the baseline attacker's, and the host layer still ranks above the service layer ($\rho = 0.03$ and -0.03)
- Untested: interactions between values

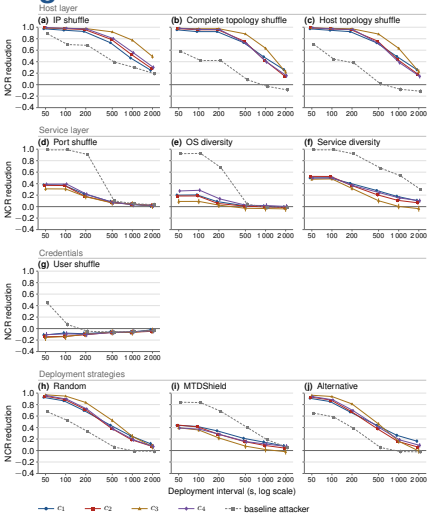
Dissertation Appendix D.1, threat T4.

Backup: what each component of the APT attacker model changes

Component removed	MTD	NCR with	without	Cohen's d
Attack profiles (use the whole attack graph)	no MTD	0.171	0.189	-0.28
	IP shuffle, 200 s	0.007	0.013	-0.47
	OS diversity, 200 s	0.159	0.174	-0.26
Failure matrix	no MTD	0.171	0.167	$+0.07$
	IP shuffle, 200 s	0.007	0.006	$+0.13$
Vulnerability memory	no MTD	0.171	0.169	$+0.04$
	service diversity, 200 s	0.109	0.106	$+0.07$

- Bold: not negligible ($|d| > 0.2$). The failure matrix changes *where* the attacker goes after a failure (back to Reconnaissance 76 to 83 % vs 8 to 10 %), not what it achieves
- Without the attack profiles, the same three host-layer mechanisms still rank first

Backup: each attack profile against the baseline attacker



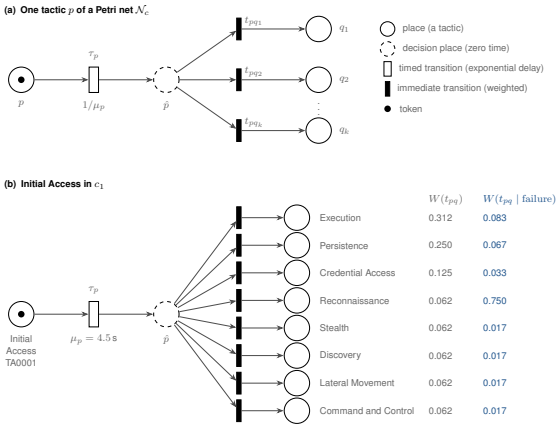
At 50 to 200 s the four attack profiles differ from one another less than the baseline attacker differs from the nearest of them (dissertation Figure 5.4).

Backup: scoring the eight properties

Attacker model	1 Pers.	2 Obj.	3 Strat.	4 Adapt.	5 Detect.	6 Cost	7 Learn.	8 Deploy.
Baseline attacker (2023)	✓	✗	part	part	✗	✗	✗	✗
Masud et al. (2025)	part	✗	✗	✗	✗	✗	✗	✗
Kim et al. (2026)	part	✗	✗	✗	part	✗	✗	✗
APT attacker model	✓	part	✓	part	part	✗	✗	✗

- Each property split into predicates; ✓ = every predicate true of the model's own behaviour; part = a behaviour predicate true, but set by a fixed rule or assumed
- One scorer, checked by blind scoring with language models (dissertation Appendix B)
- Adaptivity is in part because the failure matrix is a fixed rule of mine; detection avoidance in part because MTDSim has no IDS

Backup: the Petri net



Generalised stochastic Petri net [Ajmone Marsan 1995], extended with the failure matrix F:

$$(P, T, I, O, W, M_0, F)$$

Base weight of the move from tactic p to q:

$$W(t_{pq}) = \frac{n(p,q)}{\sum_{q'} n(p,q')}$$

with $n(p,q)$ the attack flows that draw $p \to q$. After a failed attack action:

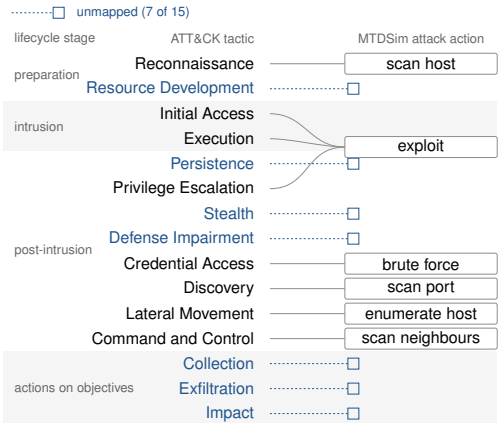
$$W(t_{pq} \mid \text{failure}) = \frac{W(t_{pq})\, F(p,q)}{\sum_{q'} W(t_{pq'})\, F(p,q')}$$

One token, starting on Reconnaissance.

Backup: tactic durations and the tactic-to-action mapping

Tactics	Mean (s)
Initial Access, Privilege Escalation, Credential Access, Lateral Movement (MTDSim's exploit)	4.5
Reconnaissance, Discovery (MTDSim's three scans)	35
Persistence, Stealth, Command and Control	45
Collection, Exfiltration, Impact	36
Execution, Defense Impairment	22.5
Resource Development	0

Last four rows: judgement, kept on the baseline attacker's time scale and under the 200 s default interval.

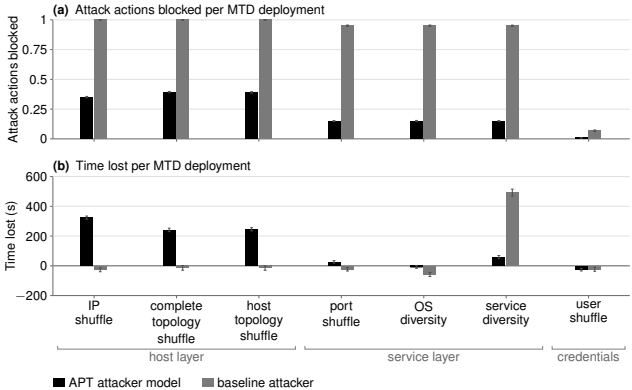


Backup: MTDShield as run

- Run as released by Tay (2024); correcting the release would make it my agent, not Tay's
- Trained at one deployment interval, against the baseline attacker, on 100-host networks
- Known limits of the release: published scores produced with exploration at one; four evaluation inputs read as zero; "do nothing" never a learning target
- Its detection input reads an unmapped tactic as "nothing detected"
- So the result bounds MTDShield *as released*, not MTDShield retrained against the APT attacker model

Dissertation Appendix E, threat T12.

Backup: disruption per MTD deployment, and ASP



- Measured at 2 000 s, because the next deployment caps time lost at the interval
- The baseline attacker's time lost is at or below zero after the first deployment: sent back to scan host, it gets a second attempt at its first host, as MTDSim intends
- On ASP reduction the layer order also holds against each attacker (Table F.3)

Backup: verification, and why no comparison with earlier MTDSim numbers

- An event tracer replays any run step by step, across Petri net, attacker and MTD
- MTDSim's code audited against what its papers describe; unit tests of the Petri net and the simulator
- Fixed seeds reproduce every run exactly
- Defects found in MTDSim were repaired before the evaluation, so results are not compared numerically with Zhang (2023) or Tay (2024)

Dissertation threat T3.

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